

SarcasmLens – Subjective Sarcasm Detection in Code-Mixed Text

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1. Problem and Motivation

The Problem Statement

To construct a system that automatically identifies sarcasm in **Code-Mixed (Hindi-English)** social media text, distinguishing subjective sarcastic expressions from literal sentiment under noisy conditions.

Key Challenges & Motivation:

- **Linguistic Complexity:** Indian social media relies heavily on Hinglish (Code-Switching), making standard English NLP models ineffective.
- **Subjectivity Cues:** Sarcasm often relies on subtle cues like exaggeration, emotive punctuation (e.g., "??!!"), and cultural context rather than explicit negative words.
- **Ambiguity:** A sentence like "*Wah modiji wah*" can be praise or sarcasm depending entirely on context and subjectivity.
- **Objective:** Move beyond simple sentiment classification to capture the *intent* behind the text.

2. Data Modelling

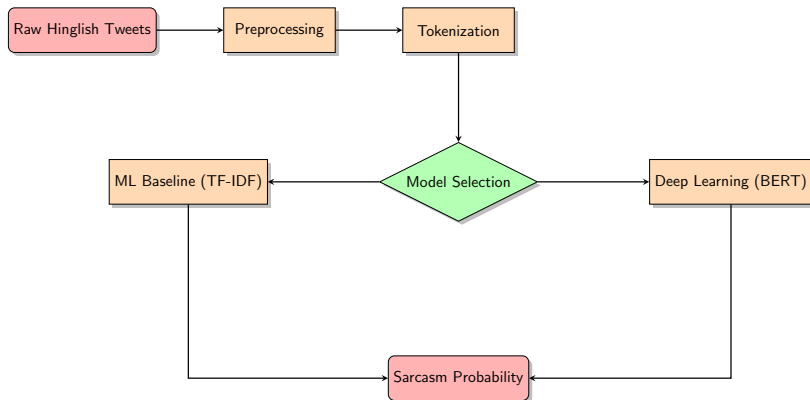
Dataset Overview (Code-Mixed Tweets):

Partition	Count	Purpose
Training Set	8,456	Learning linguistic patterns
Validation Set	2,114	Hyperparameter tuning
Test Set	2,109	Evaluation on unseen data

Preprocessing Pipeline:

- ➊ **Noise Removal:** Stripping URLs, mentions (@User), and hashtags (#) while preserving emotive punctuation crucial for sarcasm.
- ➋ **Language Preservation:** Retaining both English and Hindi characters (Unicode 0900–097F) to maintain code-mixed structure.
- ➌ **Normalization:** Lowercasing and whitespace standardization.
- ➍ **Label Mapping:** Converting subjective labels YES/NO $\rightarrow$ 1/0.

3. Methodology: Workflow



Proposed Approach:

- **Representation:** Comparative analysis between statistical n -grams (TF-IDF) and Contextual Embeddings (Transformers).
- **Strategy:** Fine-tuning pre-trained multilingual models to capture the "noisy" and informal nature of the text.

4. Model Framework & Design Choices

A. Statistical Baselines (SarcasmLens)

- **Input:** TF-IDF Vectors (8000 features, ngram 1-2).
- **Models Evaluated:**
 - Random Forest
 - Logistic Regression
 - **LinearSVC** (Selected as Best Baseline)
- **Why:** To establish a lower bound for performance using explicit word frequencies.

B. Transformer Architectures

- **Input:** Tokenized sequences (Max Len 128).
- **Models:**
 - bert-base-multilingual
 - hing-roberta (Domain Adaptive)
 - hing-mbert (Domain Adaptive)
- **Design Choice:** Using 13cube-pune models pre-trained on mixed-code corpora to better capture "Hinglish" semantics.

5. Evaluation Strategy

Evaluation Metrics

- **Weighted F1-Score:** Primary metric to balance Precision and Recall given the subjective nature of the data.
- **Accuracy:** For overall system performance.

Experimental Setup:

- **Environment:** Kaggle T4 GPU
- **Hyperparameters (DL):** Batch Size 16, Learning Rate $2e^{-5}$, 3-4 Epochs.
- **Validation Strategy:** Models saved based on best Validation F1-Score, then evaluated on the hold-out Test Set.

6. Results Insights

Table: Performance Comparison on Test Set ($N = 2109$)

Model	Accuracy	F1-Score (W)	Category
Random Forest	95.55%	0.9559	ML Baseline
Logistic Regression	96.22%	0.9622	ML Baseline
LinearSVC	96.74%	0.9674	Strongest ML
Simple mBERT	98.29%	0.9829	Multilingual
Hing-RoBERTa	98.58%	0.9858	Code-Mixed
Hing-mBERT	98.62%	0.9862	SOTA

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Note: The original log showed Logistic Regression validation F1 (0.9622), but LinearSVC was selected as best baseline for Test evaluation.

Key Findings:

- **LinearSVC** proved to be the strongest performing classical ML model.
- **Domain Adaptation:** Hing-mBERT outperformed standard mBERT, proving that pre-training on code-mixed data better captures the stylistic nuances of Indian social media sarcasm.

7. Conclusion

- **Objective Met:** The system successfully identifies sarcastic expressions with high accuracy ($> 98\%$), effectively handling code-mixing and noise.
- **Linguistic Insight:** Sarcasm in Hinglish is better represented by contextual embeddings (Transformers) than simple word counts (TF-IDF), as the "tone" often lies in the arrangement of words rather than specific keywords.
- **Design Justification:** The choice to use Hing-mBERT is justified by the empirical gain over standard multilingual models.
- **Future Work:**
 - Integrating audio/visual cues (multimodal sarcasm detection).
 - Testing on other Indic code-mixed pairs (e.g., Bengali-English).